

llmXive Automated Review of SearchSwarm: Towards Delegation Intelligence in Agentic LLMs for Long-Horizon Deep Research

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper makes several strong factual claims regarding performance benchmarks and model comparisons that require closer scrutiny of the supporting evidence and citations.

First, the claim in Section 3.2 that SearchSwarm “exceeds GPT-5.2-Thinking” on BrowseComp is problematic. Table 1 lists GPT-5.2-Thinking at 65.8 and SearchSwarm at 68.1. However, the citation `openai2025gpt52` refers to a model release in 2025, while the paper itself cites numerous 2026 models (e.g., Kimi K2.6, GLM-5). If “GPT-5.2” is a hypothetical or unreleased model, the claim of exceeding it is speculative. If it is a real model, the authors must ensure the benchmark conditions (context window, temperature, etc.) are identical, which is not explicitly detailed. The phrasing “exceeds” suggests a clear victory, but the 2.3 point margin (68.1 vs 65.8) is not accompanied by statistical significance testing.

Second, the absolute claim in Section 3.2 that “Without fine-tuning, the base model never invokes `call_sub_agent`” is a strong negative result. The text does not specify the dataset size or the number of trials used to reach this “0%” conclusion. In empirical research, “never” is a high bar; if the evaluation set was small (e.g., 10-20 questions), this claim is an overgeneralization. The authors should qualify this statement with the sample size or provide a distribution of invocation rates for the base model.

Third, the claim of “SOTA among 30B-A3B models” in the abstract and Section 3.2 relies on a very narrow margin over MiroThinker-1.7-mini (68.1 vs 67.9 on BrowseComp). While technically the highest score, the difference is 0.2 points. Without error bars or statistical significance tests, claiming “SOTA” based on such a marginal difference is potentially misleading. The text should acknowledge the closeness of the results or provide evidence that the difference is statistically significant.

Finally, the citation `openai2025gpt52` and `anthropic2025opus45` (and others with 2025/2026 dates) appear to be future-dated relative to the current real-world date. If this is a “future paper” scenario, the claims about exceeding these models are hypothetical. If the paper is intended to be a real preprint, these citations may be hallucinated or refer to non-existent models, which would invalidate the comparative claims. The authors must clarify the provenance of these baseline models.

Overall, the claims are generally supported by the provided tables, but the language used (“exceeds”, “never”, “SOTA”) is often stronger than the evidence (marginal differences, lack of statistical testing, potential hallucinated baselines) permits.

Action items.

- **[writing]** The claim that SearchSwarm ‘exceeds GPT-5.2-Thinking’ on BrowseComp (Sec 3.2) is unsupported by Table 1. The table shows 68.1 vs 65.8, but the text implies a direct comparison without clarifying if metrics or contexts differ. The phrasing ‘exceeds’ overstates

the significance of a 2.3-point margin without statistical validation.

- **[science]** The citation ‘openai2025gpt52’ refers to a ‘GPT-5.2’ model with a 2025 release date. Given the paper’s future-dated citations (e.g., 2026 for Kimi), verify if ‘GPT-5.2’ is a real, public model. If hypothetical, the claim of ‘exceeding’ it is speculative and must be qualified, not stated as an empirical result.
- **[writing]** The claim in Sec 3.2 that the base model ‘never’ invokes `call_sub_agent` is an absolute statement lacking evidence. The text does not provide the sample size or test set used to derive this 0% rate. If the evaluation set was small, this is an overgeneralization. Qualify with ‘in our evaluation of X samples’.
- **[writing]** The claim of ‘SOTA among 30B-A3B models’ (Abstract, Sec 3.2) relies on a 0.2-point margin over MiroThinker-1.7-mini (68.1 vs 67.9). Without statistical significance or error bars, claiming SOTA on such a marginal difference is misleading. Clarify if the difference is statistically significant.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The bar chart is visually clear but fails to support the caption’s specific claims regarding ‘four benchmarks’ and ‘model scale’ comparisons, as the chart lacks benchmark labels, y-axis units, and model size indicators.

Figure 2

The figure provides a clear and well-structured visual overview of the SearchSwarm architecture and execution flow that aligns with the descriptive text. However, the caption itself is incomplete, ending abruptly mid-sentence.

Figure 3

The figure presents tool usage distributions but is critically undermined by a non-descriptive caption that fails to explain the chart’s purpose. Additionally, the visualization is cluttered with a 0.1% slice that is effectively invisible.

Figure 4

The figure effectively visualizes the distribution of sub-agent calls, but the y-axis label ‘Percentage(%)’ is scientifically ambiguous for a distribution plot, and the chart title is inconsistent with the provided caption.

Action items.

- **[science]** Figure 1: The caption claims a comparison on ‘four benchmarks’, but the chart displays a single set of scores without specifying which benchmark these results correspond to or showing the other three.
- **[science]** Figure 1: The y-axis lacks a unit or label (e.g., ‘Accuracy (%)’ or ‘Score’), making the numerical values (e.g., 68.1) ambiguous.
- **[science]** Figure 1: The caption mentions ‘lightweight models of comparable scale’ and ‘larger models’, but the x-axis labels do not indicate model sizes (e.g., parameter counts), making this comparison impossible to verify visually.
- **[writing]** Figure 2: The caption text is truncated at the end (‘...multi-turn t’), cutting off the description of the subagent execution flow.
- **[fatal]** Figure 3: The caption ‘Main agent [tool_usage_four_datasets_pies_main.pdf]’ is a filename fragment and fails to describe the figure’s content (tool usage distribution across four datasets), making the figure unintelligible without external context.
- **[science]** Figure 3: The ‘BrowseComp’ and ‘BrowseComp-zh’ charts display a ‘0.1%’ slice for ‘Scholar’ that is visually indistinguishable from the white background, rendering the data point illegible.
- **[science]** Figure 4: The y-axis is labeled ‘Percentage(%)’ but the values (0-14) likely represent frequency counts or normalized density rather than a percentage of the total population, as the sum of these values across the x-axis does not equal 100%. This label is misleading for a distribution plot.
- **[writing]** Figure 4: The title ‘BrowseComp - Main Calls To Sub-Agents’ is redundant and inconsistent with the caption ‘Distribution of call_sub_agent invocation counts per question’; the title should be removed or aligned with the caption.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on specialized terminology that creates a barrier for non-specialist readers. The central concept, “delegation intelligence,” is introduced in the Abstract and

Introduction without a clear, standalone definition, assuming the reader already understands the specific nuance the authors intend. This term should be explicitly defined upon first use.

Furthermore, the paper frequently uses acronyms without expansion. “SFT” appears in Section 3.3 without being spelled out as “supervised fine-tuning.” Similarly, “SOTA” is used in Sections 4.2 and 4.6; this is informal jargon that should be replaced with “state-of-the-art” in a formal publication. The “ReAct” framework is mentioned in Section 3.1; while well-known in the field, the acronym should be expanded (Reasoning and Acting) for broader accessibility.

Finally, phrases like “token-expensive” (Section 3.2) rely on specific LLM implementation details that may not be immediately clear to all readers. Replacing such terms with more descriptive language (e.g., “computationally costly”) would improve the paper’s readability without sacrificing precision. These changes are necessary to ensure the work is accessible to the broader AI community beyond immediate specialists.

Action items.

- **[writing]** The manuscript relies heavily on specialized terminology that creates a barrier for non-specialist readers. The central concept, “delegation intelligence,” is introduced in the Abstract and Introduction without a clear, standalone definition, assuming the reader already understands the specific nuance the authors intend. This term should be explicitly defined upon first use. Furthermore, the paper frequently uses acronyms without expansion. “SFT” appears in Section 3.3 without being spelled out

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a coherent argument for “delegation intelligence” as a solution to context window limitations, and the logical flow from the problem definition to the proposed “main-distributes, sub-executes” architecture is sound. The methodology section clearly defines the ReAct formulation extended with delegation, and the training objective logically follows the data synthesis strategy.

However, there are specific logical gaps in the comparative analysis and the interpretation of ablation results:

1. **Unfair Baseline Comparison (Section 4.2):** The paper claims SearchSwarm “exceeds GPT-5.2-Thinking” on BrowseComp (68.1 vs 65.8). Table 1 indicates that SearchSwarm’s result is marked with an asterisk (*), *denoting “results with context management,” whereas the GPT-5.2-Thinking entry lacks this marker. Logically, comparing a model utilizing a specific context management strategy (delegation) against a baseline that may not be using an equivalent strategy (or is using a different one) conflates the benefit of the strategy* with the benefit of the model weights.* To support the claim that the *model* is superior, the

comparison must be apples-to-apples regarding context management capabilities, or the claim must be reframed to explicitly state that the gain is due to the delegation strategy.

2. **Ablation Causality (Section 4.3):** The ablation study attributes a +10.0 point gain to the “Full Harness” over “Tool only.” While the text lists the four principles of the harness, it does not explicitly confirm that the “Tool only” baseline was run with the exact same context window constraints (128K main / 64K sub) and temperature settings. If the “Tool only” baseline used a smaller context window or different inference parameters, the causal link between the *harness principles* and the performance gain is weakened. The logic requires that the only variable changing between the “Tool only” and “Full Harness” conditions is the prompt engineering/harness logic.
3. **Generalization Claim Nuance (Section 4.5 & Conclusion):** The conclusion states that delegation patterns “generalize to single-agent settings.” The data shows a drop from 68.1 (multi-agent) to 52.0 (single-agent) on BrowseComp. While 52.0 is an improvement over the base model (43.5), the term “generalize” can be interpreted as “maintains performance.” The logic holds that the *skill* of decomposition transfers, but the phrasing in the conclusion risks implying that the single-agent mode is nearly as effective as the multi-agent mode. Clarifying that the *capability* generalizes, even if the *efficiency* drops without the sub-agent mechanism, would be more logically precise.

These issues do not invalidate the core contribution but require clarification to ensure the conclusions strictly follow from the presented evidence.

Action items.

- **[science]** The claim that SearchSwarm ‘exceeds GPT-5.2-Thinking’ on BrowseComp (68.1 vs 65.8) is logically unsupported because the baseline result lacks the ‘*’ marker indicating context management, while the proposed method uses it. This creates an unfair comparison of capabilities rather than just model performance.
- **[writing]** The ablation study in Section 4.3 claims the ‘Full Harness’ yields a +10.0 gain over ‘Tool only’, but the text does not explicitly state whether the ‘Tool only’ baseline includes the same context window sizes (128K/64K) as the full harness, leaving the causal mechanism of the gain ambiguous.
- **[writing]** The conclusion states delegation patterns ‘generalize to single-agent’ settings, yet the single-agent results (52.0) are significantly lower than the multi-agent results (68.1). The paper needs to clarify if ‘generalization’ implies the *skill* exists (which it does) or if the *performance* is comparable, as the current phrasing risks overclaiming efficacy in the single-agent mode.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper presents a compelling architecture for delegation intelligence, but several claims overreach the empirical evidence provided in the results tables and ablation studies.

First, the assertion in Section 3.2 that the model “exceeds GPT-5.2-Thinking” is misleading. While SearchSwarm achieves 68.1 on BrowseComp compared to GPT-5.2-Thinking’s 65.8, the reverse is true on BrowseComp-ZH (73.3 vs. 76.1). The text presents this as a general victory, ignoring the metric where the larger model outperforms the proposed method. This selective framing exaggerates the model’s relative standing against frontier closed-source systems.

Second, the caption for Figure 1 claims the model “remains competitive with models over 10x larger.” This is an overgeneralization. While SearchSwarm outperforms GPT-5 on BrowseComp, it lags behind GPT-5 on BrowseComp-ZH and GAIA (76.4 vs. 82.5 is a win, but GPT-5 is not 10x larger in the same parameter count sense as the 671B DeepSeek V3.2). The comparison to DeepSeek V3.2 (671B) shows SearchSwarm trailing on BrowseComp-ZH (73.3 vs. 65.0 is a win, but DeepSeek V3.2 is 67.6 vs 68.1 on BrowseComp). The claim of being “competitive” with models 10x larger is only partially supported and requires nuance regarding which specific benchmarks and models are being compared.

Finally, the conclusion states that “Delegation patterns generalize to single-agent... tasks.” Section 3.5 demonstrates that while the model improves over the base in a single-agent setting (52.0 vs 43.5), it suffers a massive performance drop compared to its multi-agent configuration (68.1). Describing this as “generalization” without qualifying the significant loss in efficacy when the delegation mechanism is disabled overstates the transferability of the learned skills. The model appears to rely heavily on the multi-agent architecture rather than having internalized a robust single-agent reasoning capability.

Action items.

- **[science]** The claim that SearchSwarm ‘exceeds GPT-5.2-Thinking’ (Section 3.2) is an overreach. Table 1 shows SearchSwarm (68.1) only exceeds GPT-5.2-Thinking (65.8) on BrowseComp, but GPT-5.2-Thinking scores 76.1 on BrowseComp-ZH, where SearchSwarm scores 73.3. The text implies a general superiority that the data does not support across all metrics.
- **[writing]** The statement that SearchSwarm ‘remains competitive with models over 10x larger’ (Figure 1 caption) is an overgeneralization. While it beats GPT-5.2-Thinking on one metric, it trails significantly on others (e.g., GAIA: 82.5 vs. 76.4 for GPT-5, but GPT-5 is not 10x larger; the comparison to 10x larger models like DeepSeek V3.2 is mixed). The claim needs qualification to reflect specific benchmark performance rather than a blanket statement.
- **[science]** The conclusion states ‘Delegation patterns generalize to single-agent and open-ended tasks’ (Section 6). However, Section 3.5 shows a significant performance drop in the single-agent setting (52.0 vs 68.1 on BrowseComp). While it improves over the base model, claiming ‘generalization’ without acknowledging the substantial performance degradation in the absence of subagents overstates the robustness of the learned behavior.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper presents a multi-agent system (SearchSwarm) designed for long-horizon deep research, utilizing web browsing tools and delegated sub-agents. From a safety and ethics perspective, the primary concerns revolve around data privacy, the potential for generating hallucinated citations in a research context, and the dual-use nature of autonomous information gathering.

Data Privacy and Consent: In Section 3.3 (“Supervised Fine-tuning”), the authors state that data is sourced from RedSearcher and OpenSeeker and that they will “open-source our harness, weights, and data.” While the source datasets are external, the authors must explicitly confirm in the manuscript that the synthesized trajectories used for training do not contain Personally Identifiable Information (PII) or private user queries from the original datasets. Given the nature of “deep research,” there is a risk that user prompts could contain sensitive personal or corporate information. A statement confirming that a privacy filter was applied to the training data before release is necessary to ensure compliance with data protection norms.

Hallucination and Misinformation: The system is designed to produce “citation-grounded reports” (Section 3.2). However, the case study in Appendix C (Section 5) demonstrates the model synthesizing complex facts about real-world infrastructure projects (e.g., Coomera Connector, Inland Rail) and specific corporate joint ventures. While the example shows correct retrieval, the reliance on LLMs to synthesize these reports carries a risk of “citation hallucination,” where the model invents plausible-sounding but non-existent sources or misattributes facts. The paper should include a discussion on the robustness of the citation verification mechanism and the potential for the system to be used to generate misleading research reports if the verification step is bypassed or fails.

Dual-Use and Autonomous Browsing: The system employs autonomous web browsing tools (`search`, `visit`, `google_scholar`) to gather information. While intended for research, this capability could be misused for automated scraping of restricted content, social engineering (by gathering personal details), or generating disinformation campaigns at scale. The authors should briefly address these dual-use risks in the “Limitations” or “Ethical Considerations” section, perhaps noting that the system is intended for benign research tasks and that the authors have not tested it against adversarial prompts designed to extract sensitive data.

Conclusion: The paper does not appear to violate fundamental ethical principles, but it lacks explicit statements regarding data privacy in the released artifacts and the potential risks of autonomous information gathering. Addressing these points will strengthen the paper’s ethical standing.

Action items.

- **[writing]** The paper presents a multi-agent system (SearchSwarm) designed for long-horizon deep research, utilizing web browsing tools and delegated sub-agents. From a safety and

ethics perspective, the primary concerns revolve around data privacy, the potential for generating hallucinated citations in a research context, and the dual-use nature of autonomous information gathering. Data Privacy and Consent: In Section 3.3 (“Supervised Fine-tuning”), the authors state that data is sourced from RedSearcher a

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling architecture for “delegation intelligence” and reports strong performance gains on deep research benchmarks. However, the scientific evidence supporting the magnitude and robustness of these claims is currently insufficient due to a lack of statistical rigor.

The primary concern lies in the reporting of quantitative results in Section 4.2 and Table 1. The authors claim SearchSwarm achieves state-of-the-art results among lightweight models, citing specific margins (e.g., 68.1 vs. 67.9 on BrowseComp). In the absence of reported standard deviations, confidence intervals, or results from multiple random seeds, these point estimates are scientifically weak. A 0.2-point difference is well within the noise floor of typical LLM evaluation pipelines. Without statistical significance testing (e.g., t-tests or bootstrap confidence intervals), the claim that SearchSwarm strictly outperforms MiroThinker is not empirically supported.

Furthermore, the ablation study in Section 4.3 relies on a 200-question subset of BrowseComp to demonstrate the harness’s effectiveness (+10.0 gain). The manuscript does not specify the sampling strategy for this subset. If the subset was not randomly selected, the results may suffer from selection bias, inflating the perceived impact of the method. To ensure the evidence is robust, the authors must clarify the sampling method or, preferably, report ablation results on the full benchmark with variance metrics.

Finally, the comparison against closed-source models (e.g., GPT-5.2-Thinking) is presented as a definitive superiority claim. While the point estimate is higher, the lack of error bars on the authors’ own results makes it difficult to assess the reliability of this advantage. The paper should include a discussion on the statistical significance of these gains or explicitly frame them as preliminary observations pending further variance analysis.

Action items.

- **[science]** Section 4.2 (Main Results) and Table 1 lack statistical significance testing. The margin between SearchSwarm (68.1) and MiroThinker (67.9) on BrowseComp is 0.2 points. Without reported standard deviations, confidence intervals, or p-values from multiple runs, it is impossible to determine if this is a genuine improvement or noise.
- **[science]** The ablation study in Section 4.3 (‘Effectiveness of the Harness’) uses a 200-question subset but does not specify if this subset is a random sample or a curated hard/easy set.

If not random, the +10.0 gain may be biased. Please clarify the sampling methodology or provide results on the full benchmark.

- **[science]** The claim that SearchSwarm ‘exceeds GPT-5.2-Thinking’ (Section 4.2) relies on a single-point comparison (68.1 vs 65.8) without error bars. Given the proprietary nature of the baseline, the authors must explicitly state the variance of their own model’s performance across seeds to validate the robustness of this claim.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The manuscript presents a compelling architecture for delegation intelligence, but the statistical rigor of the empirical evaluation requires strengthening to support the claims of state-of-the-art performance.

First, the primary results in Table 1 (Section 4.2) report point estimates for benchmark scores (e.g., 68.1 vs. 67.9 on BrowseComp) without any measure of variance. Given that the performance gaps between SearchSwarm and the closest competitors (MiroThinker-1.7-mini) are often less than 1.0 point, it is impossible to determine if these differences are statistically significant or within the noise of the evaluation process. The authors must report standard deviations (calculated over multiple random seeds or bootstrap samples) or perform significance testing (e.g., paired t-tests or Wilcoxon signed-rank tests) to validate that the observed improvements are not due to chance.

Second, the ablation study in Section 4.3 (“Effectiveness of the Harness”) relies on a 200-question subset of BrowseComp. The text does not specify the sampling method (e.g., random stratified sampling) or whether the evaluation was repeated across multiple seeds. Without this information, the reported +10.0 gain could be an artifact of a favorable subset selection. The authors should clarify the sampling protocol and ideally report the variance of the gain across different subsets or seeds.

Finally, in Section 4.5, the paper claims generalization to a single-agent setting by disabling the `call_sub_agent` tool. However, it compares the fine-tuned model in this mode against the base model’s standard performance. To properly isolate the effect of the fine-tuning on decomposition skills, the base model should also be evaluated in the same single-agent configuration (with the tool disabled) to provide a fair baseline. The current comparison conflates the effect of the architectural constraint with the effect of the training data.

Addressing these statistical gaps is essential to substantiate the claim that SearchSwarm achieves a genuine, reproducible improvement over existing methods.

Action items.

- **[science]** Report statistical significance (e.g., p-values, confidence intervals, or standard

deviations) for the benchmark comparisons in Table 1. The marginal gains (e.g., 68.1 vs 67.9 on BrowseComp) are too small to claim superiority without variance estimates or significance testing across multiple runs.

- **[science]** Clarify the experimental protocol for the ablation study in Section 4.3. Specify if the 200-question subset was sampled randomly and if the base, tool-only, and full-harness models were evaluated on the exact same seed set to ensure the +10.0 gain is not due to sampling variance.
- **[science]** The ‘Generalization to Single-Agent’ results (Section 4.5) lack a baseline comparison for the specific single-agent configuration. Clarify if the base model was also evaluated in a single-agent mode to isolate the effect of the fine-tuning versus the architectural change.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a strong command of technical vocabulary and generally maintains a professional academic tone. The logical flow from the problem statement (context limits) to the proposed solution (delegation intelligence) is clear and well-structured. However, several sentence-level issues and structural inconsistencies detract from the overall readability and polish of the text.

In Section 2.1 (Formulation), the definition of the trajectory H_T is presented as a mathematical fragment (“A trajectory $H_T = \dots$ ”) rather than a complete sentence. This breaks the grammatical flow and should be corrected to “A trajectory H_T is defined as...” or similar. Additionally, the Introduction’s contribution list suffers from a lack of parallel structure; the first bullet uses “We explore” while the others use “We synthesize” and “We open-source.” Aligning these verbs would improve the rhythmic consistency of the claims.

Transitions between results and analysis could be smoother. For instance, in Section 3.2, the observation that the base model never invokes the delegation tool appears as an isolated sentence following the performance table. A brief transitional phrase linking this behavioral observation to the necessity of the fine-tuning pipeline would enhance cohesion. Finally, the Appendix Case Study (Section A.3) exhibits inconsistent formatting, switching between narrative paragraphs and bulleted lists for the “Rejected alternatives” section. Standardizing this layout would ensure visual consistency with the main body of the paper. Addressing these minor grammatical and structural points will significantly elevate the manuscript’s readability.

Action items.

- **[writing]** In Section 2.1 (Formulation), the trajectory definition ‘ $H_T = (q, (a_0, o_0), \dots, (a_T, o_T), y)$ ’ lacks a verb and reads as a fragment. Rewrite as a complete sentence, e.g., ‘A trajectory H_T is defined as...’.

- **[writing]** The Introduction lists three bullet points but inconsistently mixes ‘We explore’ (present tense) with ‘We synthesize’ and ‘We open-source’. Ensure parallel structure and consistent tense across all contribution statements.
- **[writing]** In Section 3.2 (Main Results), the sentence ‘Without fine-tuning, the base model never invokes `call_sub_agent`’ appears abruptly. Add a transitional phrase to connect this observation to the preceding performance comparison.
- **[writing]** The Appendix Case Study (Section A.3) uses a mix of narrative paragraphs and bulleted lists for the ‘Rejected alternatives’ section. Standardize the formatting to ensure visual consistency with the rest of the paper.